



GE Medical Systems

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**Service Introduction Planner (iSERV)
Mobile LightSpeed**

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REVISION HISTORY

Revision History

Revision	Date	Reason for change
0	10/18/01	Initial Release

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SERVICE INTRODUCTION PLANNER (ISERV)

OVERVIEW

This Introduction Planner provides information on the LightSpeed Plus product to be installed in non-fixed site environments.

This planner is intended to provide field service personnel with information regarding the LightSpeed CT non-fixed site system going into forward production. This Service Introduction Planner is NOT a service manual, direction, or user guide. It is a collection of information taken from several source documents, ALL of which are available and should be used to install, maintain, and operate the CT LightSpeed Plus non-fixed site system.

A non-fixed environment is any prefabricated, stand-alone environment, which is designed to hold a CT scanner. Non-fixed environments are shipped with the CT scanner preinstalled prior to the environment/scanner package being delivered to the customer site. Examples of non-fixed environments are mobile trailers, transportable trailers, relocatable buildings, and ISO containers.

The LightSpeed Plus system for non-fixed environments is very similar to that for fixed site installations. Minor changes have been incorporated into the non-fixed version to accommodate the mobile environment needs (vibration, multi-hospital users on the same equipment, etc.)

1.1 - PRODUCT STRUCTURE

Product Name	Engineering Program Name	Catalog Number	Description
Non-fixed LightSpeed	Helios 2.1 Mobile	B7850MB	Contains the following: B7850NG GANTRY B7850NP PDU B7850NT TABLE B7850SC SHIP COLLECTOR B7850TE ENGLISH LANGUAGE B7800KE ENGLISH KEYBOARD B7850NB PLUS CONSL,FLAT B7850LL LONG CABLE B7999PY UPS+HW KIT, MOBILE B7850MK LTSPD MOBILE KIT

1.2 - SYSTEM SPECIFICATIONS

The Non-fixed LightSpeed Plus configuration will have the same system performance specifications as the equivalent fixed site LightSpeed Plus. All options specified in Non – Fixed CT LS system spec & Install requirements would be available for LightSpeed Plus non-fixed site installations with the following exceptions:

1.3 - KEY FEATURES OF NON-FIXED OFFERINGS

1.3.1 SERVICE FEATURES

The service features for the non-fixed version are virtually identical to the H2.1 fixed site features. There are no changes to the diagnostics or service procedures. Software changes are documented elsewhere within this iSERV.

1.3.2 CUSTOMER FEATURES

The customer features of the non-fixed LightSpeed Plus systems are the same as for the fixed site equivalent. There are some additional features built into the customer software which will accommodate multi-hospital users on the same scanner, a modified shut down procedure which automatically tilts the gantry back for transport, the addition of flat panel monitors as standard monitors in non-fixed environments, and hardware to support the table for transport.

1.4 - SERVICE AND OPERATOR DOCUMENTATION

All service documents are available at the CT Service Engineering Web-site
http://www.ct.med.ge.com/tech_pubs/welcome.htm

Direction	Description
2243311-100	Pre-Installation/Site Planning Manual, Rev 03
2243312-100	Installation Manual, Rev 02
2243314-100	System Service Manual, Rev 05
2243316-100	Illustrated Parts Manual , Rev 04
2243317-100	Schematics , Rev 02
2243319-100	SW Load Procedure , Rev 06
2243321-100	Applications Tips and Work-Arounds, Rev 05
2281207-100	Technical Reference Manual, Rev 02
2281208-100	Learning and Reference Guide, Rev 03
2141834-100	CT PM Procedures, Rev 10
????	Mobile Start Up /Shutdown
2288566-100	Prospective Gating - LightSpeed Plus, Rev 02
2315243-100	UPS Installation Mobile LightSpeed
2305587-100	Non-Fixed CT LS Systems Spec & Install Requirements, Rev 00
2305588-100	Non-Fixed CT LS Mechanical Integration Requirements, Rev 00
2318573-100	Non-Fixed CT LS Electrical Integration Procedures, Rev 00
2318669-100	On-Site Mobile Set-up Procedure, Rev 00
2211224-100	LightSpeed Series Options Installation, Rev 01

1.5 - SUMMARY OF HARDWARE (MAIN DIFFERENCES VS. LIGHTSPEED FIXED SITE)

Customer-Driven User Interface Features

Flat panel monitors as standard

Light on van exterior to identify if the gantry is ready for transport (tilted back fully), HVAC ok, 480/400AC ok.

2.0 - INTEGRATION

Mechanical and Electrical integration requirement for the non-fixed environment manufacturer can be found in the Non-Fixed CT LS Mechanical Integration requirement ref: 2305588-100 Rev 00 and Non-Fixed CT LS Electrical Integration Procedures, 2318573-100 rev 00

The respective environment manufacturer provides site-specific pre-installation information. This information will include the pad requirements, power requirements and connector requirements, phone and network requirements, utility requirements (water, drains, etc. as required), and radiation safety zones as required at a minimum.

3.0 - INSTALLATION

3.1 - INSTALLATION TIME

Turnover time at customer delivery will be no longer than 6 hours at the customer site. There will be no further service involvement at subsequent customer locations. Power ground audit is required for all sites.

3.2 - DOCUMENTS & TOOLS

Changes required to the LightSpeed Plus Service documentation and Operators documentation will be made to the existing fixed site operators and service documentation. Separate documents will be developed for the non-fixed Mechanical, Electrical and installation requirements for non-fixed environment. The part numbers are as follow:

2305588-100 Non-Fixed CT LS Mechanical Integration Requirements

2318573-100 Non-Fixed CT LS Electrical Integration Procedure

2305587-100 NON-Fixed CT LS Systems Spec & Install Requirements

Gantry service tools are supplied in the install support kit. Mobile Gantry dollies may be different than fixed site dollies.

SUPPLIED BY ENVIRONMENT MANUFACTURER

F.E storage cabinet provided by the Van Company. The following tools will be stored in the Cabinet

- QA Phantom (Water Filled)
- 20cm Phantom
- 48cm Phantom
- Phantom Holder

- Gantry Display
- FE Box Purple
- Rear Cover dollies
- Front Cover Dollies
- Install support kit (Box)
- Tube Hoist Kit
- Gantry Control Pad

Required Tools:

Items supplied by Field - standard tool kit as required to service the current LightSpeed Plus fixed site systems.

Items supplied by the Environment Manufacturer - any special tools or dollies required to support the van and/or the LightSpeed Plus system

Some van models may require a cabinet to be removed for major changes

3.3 - ON-SITE SET UP PROCEDURE

On - Site set up procedure can be found in On-Site Mobile Set-up Procedure, Rev 00 Ref. 2318669-100

3.4 - INSITE CONNECTION

InSite connection is mandatory on this system.

MODEM : Global modem to be shipped with each system

Standard weather resistance cables are supplied with each van

3.5 - SOFTWARE

Mobile installations of LightSpeed plus will take advantage of a software patch made specifically for mobile applications. This patch software will be added to the H2.1 software as a patch load and incorporated into H2.2 software. The contents of this patch will support for multi-hospital users on the same equipment. Specifically:

Storing multiple site IP information for networking and printing

Multiple configuration info for remote connectivity

Special shutdown process which tilt the gantry fully back to its stops for transport

This software will be configurable at set-up.

4.0 - SERVICE STRATEGY

4.1 - SERVICE RESPONSIBILITIES

Reference 2318669-100 On-Site Mobile set up procedure

4.2 - MAINTENANCE

Same as fixed-site

4.3 - PLANNED MAINTENANCE

- Maintenance Frequency depends on Site activity.

Source pole recommendations are:

- o In warranty: 4 PM visits/year
- o Out of warranty: 1PM per 250 Kslices

Maintenance Time

- o In warranty: 13 hours/year
- o Out of warranty:

- Required Tool : Items supplied by Field (Preliminary List)

Service Tool requirements will be the same as for the LightSpeed Plus fixed site product. Any special tools (dollies, etc) that are required to support the scanner in a non-fixed environment will be supplied by the environment manufacturer.

- PM Manual : 2141834

4.4 - CORRECTIVE MAINTENANCE

Correct maintenance strategy (tools, process, times, etc) for the non-fixed site version will be the same as for the fixed site LightSpeed Plus.

Major Field Replaceable Units (FRU) are

- X-Ray Tube
- High Voltage Tank
- Detector
- Slip Ring
- DAS

4.5 - SPARE PARTS – ADDITIONS FOR MOBILE APPLICATION

Maintenance policy is stock at FRU level.

Part #	Description
Gantry	
2303589	Harness, tilt switch Assy.
2303601	Harness, ct mobile tilt
Table	
46-296317p4	OBC-STC & Table EMC power supply
2302366	Pogo Stick
Console	
2306872	18.1" Global Standard LCD Monitor
PDU	

2310005	HARDENED UPS FOR MOBILE LIGHTSPEED +
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5.0 - SOFTWARE

- System software CD-ROM 2319688 includes patch for Mobile.
- The operating system and Insite Software (Class M) are same as Fixed-site
- Option key MOD will be shipped with the system

6.0 - TRAINING

There will be no specific training class for the LightSpeed Plus in non-fixed site installations. There will be a T-CON with primary F.E to discuss the differences between the Fixed site and Non-Fixed site services.



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